Ruiyi Yang

CONTACT INFORMATION
Room 369, No.6 Science Building.
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EMPLOYMENT

Shanghai Jiao Tong University 2026–present
Institute of Natural Sciences and School of Mathematical Sciences
Tenure-Track Associate Professor

Institute for Mathematical and Statistical Innovation Fall 2025
Visiting Research Scholar

Princeton University 2022-2025
Program in Applied and Computational Mathematics
Postdoctoral Research Associate. Supervisor: Amit Singer

EDUCATION

University of Chicago 2017-2022
Ph.D. in Computational and Applied Mathematics. Advisor: Daniel Sanz-Alonso

University of California, Los Angeles 2013-2017
B.S. in Mathematics (Honors)

PUBLICATIONS AND PREPRINTS

(Authors are ordered alphabetically in all papers.)

1. Yunpeng Shi, Amit Singer, Jinxin Wang, and Ruiyi Yang. Orientation Estimation of Cryo-EM Images for Molecules with Arbitrary Symmetry by Semidefinite Programming. *arXiv preprint*, 2026.
2. Anna Little, Daniel Sanz-Alonso, Mikhail Sweeney, and Ruiyi Yang. Functional Multi-Target Detection via Bispectrum Inversion. *arXiv preprint*, 2026.
3. Wenwen Li, Daniel Sanz-Alonso, and Ruiyi Yang. Bayesian Optimization on Networks. *Journal of Computational Physics*, 558:114819, 2026.
4. Bryon Aragam and Ruiyi Yang. Model-free Estimation of Latent Structure via Multiscale Nonparametric Maximum Likelihood. *arXiv preprint*, 2025.
5. Daniel Sanz-Alonso and Ruiyi Yang. Gaussian process regression under computational and epistemic misspecification. *SIAM Journal on Numerical Analysis*, 63(2):495-519, 2025.
6. Amit Singer and Ruiyi Yang. Alignment of density maps in Wasserstein distance. *Biological Imaging*, 4:e5, 2024
7. Hwanwoo Kim, Daniel Sanz-Alonso, and Ruiyi Yang. Optimization on manifolds via graph Gaussian processes. *SIAM Journal on Mathematics of Data Science*, 6(1):1–25, 2024
8. Bryon Aragam and Ruiyi Yang. Uniform consistency in nonparametric mixture models. *The Annals of Statistics*, 51(1):362–390, 2023
9. Nicolás Garcia Trillos, Daniel Sanz-Alonso, and Ruiyi Yang. Mathematical foundations of graph-based Bayesian semi-supervised learning. *Notices of the American Mathematical Society*, 69(10), 2022
10. Daniel Sanz-Alonso and Ruiyi Yang. Finite element representations of Gaussian processes: Balancing numerical and statistical accuracy. *SIAM/ASA Journal on Uncertainty Quantification*, 10(4):1323–1349, 2022
11. Daniel Sanz-Alonso and Ruiyi Yang. Unlabeled data help in graph-based semi-supervised learning: a Bayesian nonparametrics perspective. *Journal of Machine Learning Research*, 23(97):1–28, 2022
12. Daniel Sanz-Alonso and Ruiyi Yang. The SPDE approach to Matérn fields: Graph representations. *Statistical Science*, 37(4):519–540, 2022
13. John Harlim, Daniel Sanz-Alonso, and Ruiyi Yang. Kernel methods for Bayesian elliptic inverse problems on manifolds. *SIAM/ASA Journal on Uncertainty Quantification*, 8(4):1414–1445, 2020

14. Nicolás García Trillos, Daniel Sanz-Alonso, and Ruiyi Yang. Local regularization of noisy point clouds: Improved global geometric estimates and data analysis. *Journal of Machine Learning Research*, 20(136):1–37, 2019

Awards

- Harper Dissertation Fellowship, University of Chicago. 2021
  In recognition of record or achievement and professional promise, one of University of Chicago's highest honors.
- SIAM Travel Award 2020, 2021, 2022

Talks

- East Asia Section of SIAM Annual Meeting, Jeju Island, Aug 2026.
- Joint Statistical Meetings, Boston, Aug 2026
- Scientific Computing and Differential Equations, Edinburgh, Jul 2026.
- SIAM Conference on Mathematics of Data Science, Atlanta, Oct 2024.
- University of California Davis MADDD Seminar, Feb 2024.
- IMS Young Mathematical Scientists Forum—Applied Mathematics, Singapore, Jan 2024
- New Jersey Institute of Technology Statistics Seminar, Mar 2023.
- SIAM Conference on Mathematics of Data Science, San Diego, Sep 2022.
- IMSI Workshop on Expressing and Exploiting Structure in Modeling, Theory, and Computation with Gaussian Processes, Chicago, Aug 2022.
- SIAM Conference on Uncertainty Quantification, Atlanta, Apr 2022.
- Joint Statistical Meetings, Virtual, Aug 2021.
- SIAM Conference on Computational Science and Engineering, Virtual, Mar 2021.
- University of Wisconsin-Madison Statistics Seminar Virtual, Feb 2021.
- SIAM Conference on Mathematics of Data Science Virtual, Jun 2020.

Teaching Experience

- Shanghai Jiao Tong University
  - STAT 6001 Basic Mathematical Statistics. Fall 2026
- Princeton University
  - MAT321/APC321 Numerical Analysis and Scientific Computing. Fall 2023
- University of Chicago
  - CAAM 31440: Applied Analysis, CAAM 31210: Applied Functional Analysis, 2019, STAT 24300: Numerical Linear Algebra, CAAM 31511: Monte Carlo Simulation, STAT 31700: Introduction to Probability Models, CAAM 31450: Applied Partial Differential Equations, CAAM 31220: Partial Differential Equations.

Professional Services

- Conference Session Organizer
  SIAM Conference on Mathematics of Data Science, Atlanta GA. Oct 2024
  Minisymposium: "Recent Advances in Gaussian Process and Kernel Methods" (with Daniel Sanz-Alonso).
- Journal Referee
  Annals of Statistics; Journal of the Royal Statistical Society Series B; Journal of the American Statistical Association; Journal of Machine Learning Research; Statistics and Computing; Journal of Machine Learning; Journal of Computational and Applied Mathematics; IISE Transactions; Signal Processing
- Conference Reviewer
  NeurIPS; ICML; ICLR; AAAI; AISTATS